

DevelopersHub Corporation

Performance Testing Report

Week 4 – Load, Stress & Spike Test Results

Application: automationexercise.com

Tool: Apache JMeter 5.6.3

Date: May 24, 2026

Prepared by: Kashan Karim Intern – DevelopersHub Corporation

Report Version: 1.0

1. Objective

The objective of this performance test was to evaluate the responsiveness, stability, and scalability of **automationexercise.com** under simulated concurrent user loads. Three test scenarios were executed using Apache JMeter 5.6.3: a Load Test (100 users), a Stress Test (200 users), and a Spike Test (150 users with rapid ramp-up). Results were measured against predefined KPIs including response time, error rate, and throughput.

2. Test Environment

Parameter	Details
Testing Tool	Apache JMeter 5.6.3
Operating System	Windows 10
Target Application	https://automationexercise.com
Pages Tested	Home, Login (POST), Products, Signup, Contact Us
Test Scenarios	TS-01: Load (100u), TS-02: Stress (200u), TS-03: Spike (150u)

Test Date	May 24, 2026
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3. Test Scenarios Executed

Each test scenario executed the following 5-step user workflow:

- GET / → Home Page
- POST /login → Login form submission
- GET /products → Products listing page
- GET /signup → Signup page
- GET /contact_us → Contact Us page

ID	Scenario	Virtual Users	Ramp-Up Period	Loop Count	Total Samples
TS-01	Load Test	100 users	60 seconds	3 loops	1,500
TS-02	Stress Test	200 users	60 seconds	2 loops	2,000
TS-03	Spike Test	150 users	5 seconds	1 loop	750

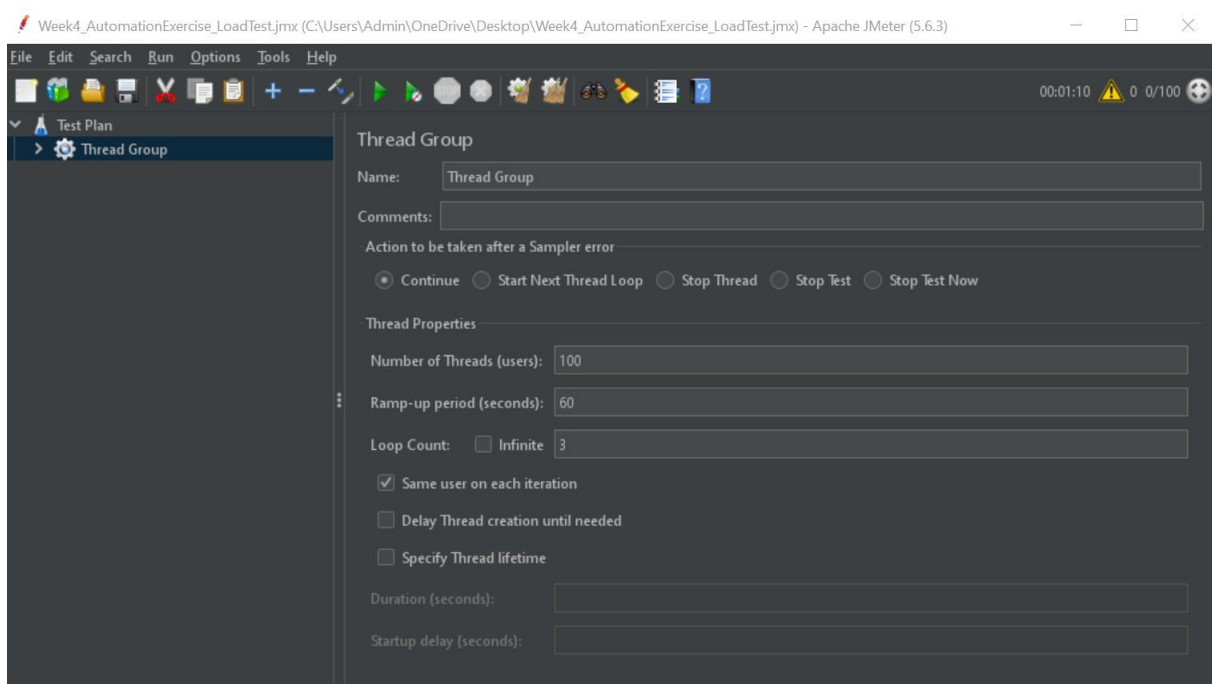
3. Test Scenarios Executed

Observation	TS-01 Load (100u)	TS-02 Stress (200u)	TS-03 Spike (150u)
Total Samples	1,500	2,000	750
Overall Avg Response Time	735 ms	409 ms*	1,506 ms
GET Home Avg	902 ms	481 ms	1,819 ms
POST Login Avg	668 ms	372 ms	1,466 ms
GET Products Avg	758 ms	451 ms	1,613 ms
GET Signup Avg	684 ms	371 ms	1,400 ms
GET Contact Avg	661 ms	372 ms	1,230 ms
POST Login Error Rate	100%	100%	100%
Overall Error Rate	36.60%	20.00%	38.93%
Throughput (req/sec)	21.9/sec	31.1/sec	59.3/sec
Max Response Time	5,056 ms	1,608 ms	3,785 ms

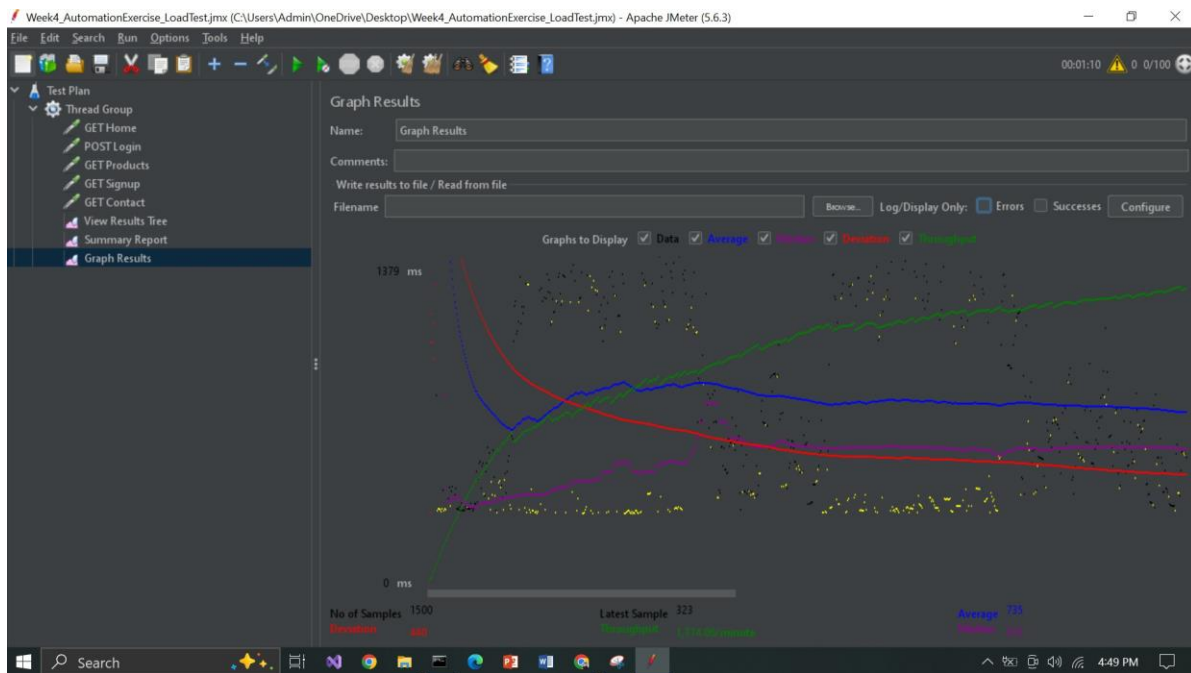
5. Graphs & Results (JMeter Screenshots)

5.1 – TS-01: Load Test (100 Users, 60s Ramp-up, 3 Loops)

Thread Group Configuration:



Graph Results – Response time over test duration:



Summary Report – Actual JMeter results:

Week4_AutomationExercise_LoadTest.jmx (C:\Users\Admin\OneDrive\Desktop\Week4_AutomationExercise_LoadTest.jmx) - Apache JMeter (5.6.3)

File Edit Search Run Options Tools Help

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Test Plan

- Thread Group
 - GET Home
 - POST Login
 - GET Products
 - GET Signup
 - GET Contact
 - View Results Tree
 - Summary Report**
 - Graph Results

Summary Report

Name: Summary Report

Comments:

Write results to file / Read from file

Filename: Browse... Log/Display Only: ☐ Errors ☐ Successes

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB...	Sent KB/sec	Avg. Bytes
GET Home	300	902	309	5056	609.04	19.67%	4.5/sec	190.47	0.82	43183.9
POST Login	300	668	309	2010	369.89	100.00%	4.8/sec	18.33	1.09	3875.2
GET Products	300	758	305	1915	389.35	21.33%	4.8/sec	183.13	0.64	38825.1
GET Signup	300	684	307	1764	370.94	21.67%	4.8/sec	30.03	0.63	6373.1
GET Contact	300	661	310	1683	361.98	20.33%	4.8/sec	38.59	0.64	8227.1
TOTAL	1500	735	305	5056	440.25	36.60%	21.9/sec	429.83	3.52	20096.9

☐ Include group name in label? ☒ Save Table Header

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View Results Tree – Pass/Fail per request:

Week4_AutomationExercise_LoadTest.jmx (C:\Users\Admin\OneDrive\Desktop\Week4_AutomationExercise_LoadTest.jmx) - Apache JMeter (5.6.3)

File Edit Search Run Options Tools Help

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Test Plan

- Thread Group
 - GET Home
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 - GET Signup
 - GET Contact
 - View Results Tree**
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View Results Tree

Name: View Results Tree

Comments:

Write results to file / Read from file

Filename: Browse... Log/Display Only: ☐ Errors ☐ Successes

Search: Case sensitive ☐ Regular exp. ☐ Search

Text

- GET Products (Pass)
- GET Contact (Pass)
- POST Login (Fail)
- POST Login (Fail)
- GET Contact (Pass)
- GET Products (Pass)
- GET Home (Pass)
- GET Signup (Pass)
- GET Contact (Pass)
- GET Signup (Pass)
- GET Contact (Pass)
- GET Signup (Fail)
- GET Products (Fail)
- GET Products (Fail)
- GET Home (Pass)
- GET Home (Pass)

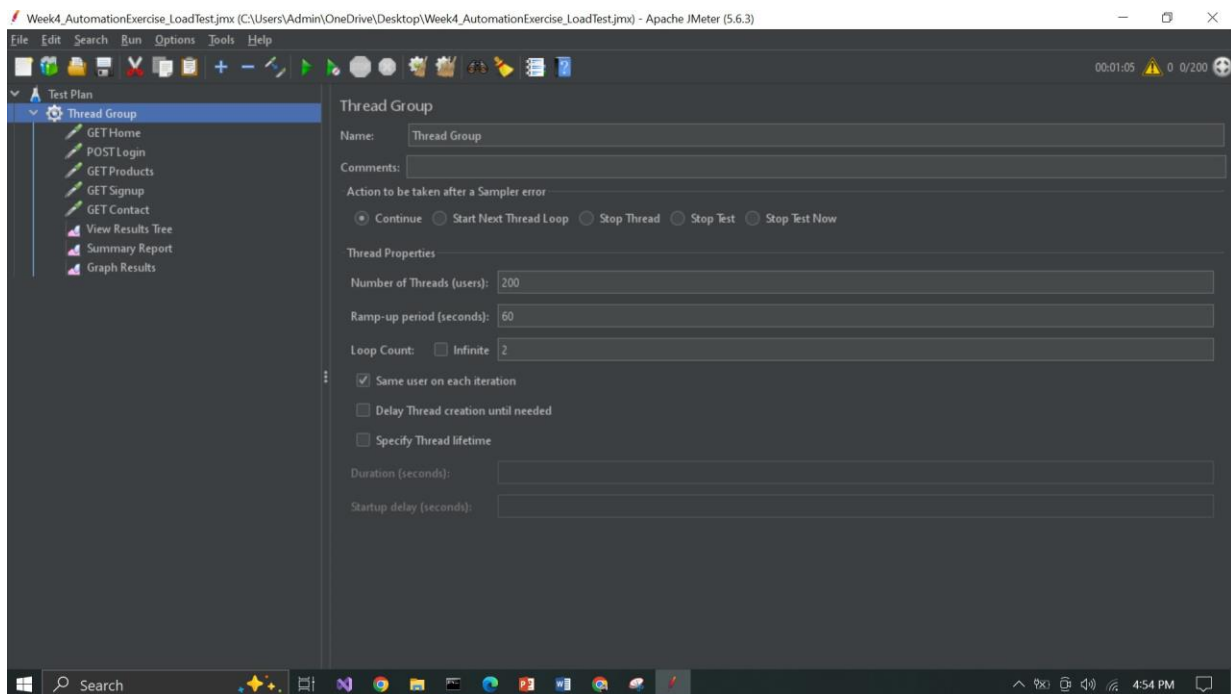
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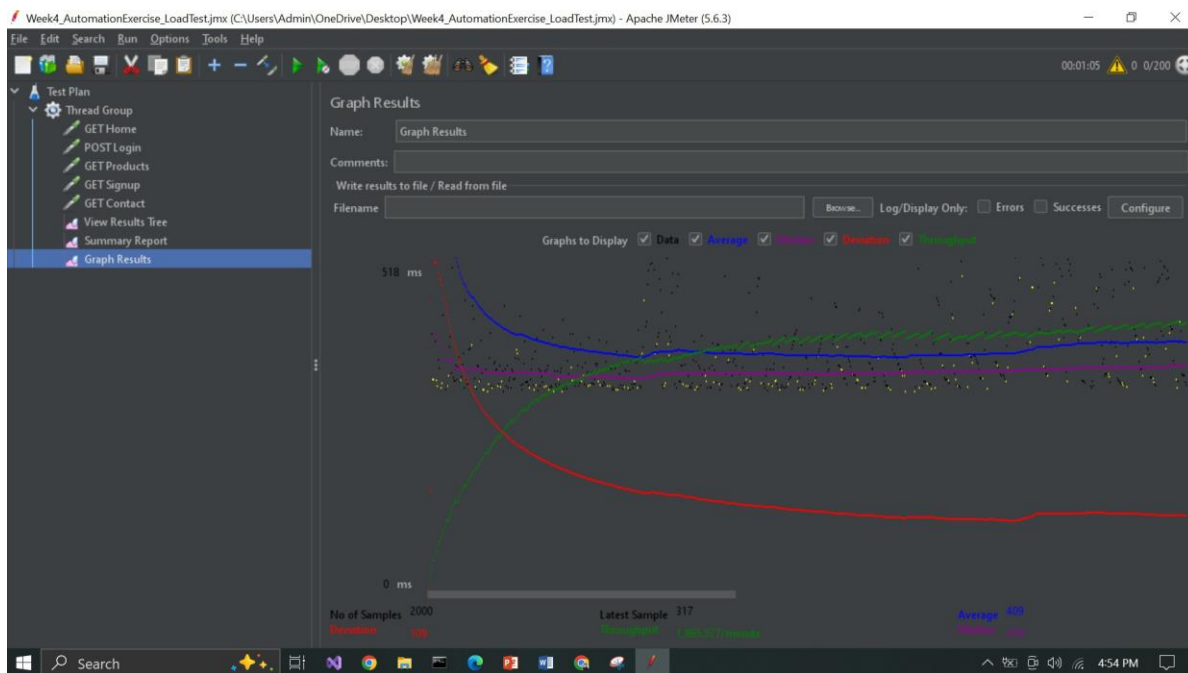
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5.2 – TS-02: Stress Test (200 Users, 60s Ramp-up, 2 Loops)

Thread Group Configuration:



Graph Results – Response time over test duration:



Summary Report – Actual JMeter results:

Week4_AutomationExercise_LoadTest.jmx (C:\Users\Admin\OneDrive\Desktop\Week4_AutomationExercise_LoadTest.jmx) - Apache JMeter (5.6.3)

File Edit Search Run Options Tools Help

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Test Plan

- Thread Group
 - GET Home
 - POST Login
 - GET Products
 - GET Signup
 - GET Contact
 - View Results Tree
 - Summary Report**
 - Graph Results

Summary Report

Name: Summary Report

Comments:

Write results to file / Read from file

Filename: Browse... Log/Display Only: ☐ Errors ☐ Successes

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB...	Sent KB/sec	Avg. Bytes
GET Home	400	481	337	1608	146.96	0.00%	6.4/sec	335.04	1.17	53515.7
POST Login	400	372	308	748	75.15	100.00%	6.5/sec	30.04	1.47	4707.9
GET Products	400	451	331	896	102.81	0.00%	6.5/sec	312.24	0.86	49087.8
GET Signup	400	371	310	697	73.06	0.00%	6.5/sec	49.93	0.84	7864.6
GET Contact	400	372	309	712	74.37	0.00%	6.5/sec	63.76	0.87	10076.7
TOTAL	2000	409	308	1608	109.38	20.00%	31.1/sec	760.62	5.00	25050.6

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View Results Tree – Pass/Fail per request:

Week4_AutomationExercise_LoadTest.jmx (C:\Users\Admin\OneDrive\Desktop\Week4_AutomationExercise_LoadTest.jmx) - Apache JMeter (5.6.3)

File Edit Search Run Options Tools Help

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Test Plan

- Thread Group
 - GET Home
 - POST Login
 - GET Products
 - GET Signup
 - GET Contact
 - View Results Tree**
 - Summary Report
 - Graph Results

View Results Tree

Name: View Results Tree

Comments:

Write results to file / Read from file

Filename: Browse... Log/Display Only: ☐ Errors ☐ Successes

Search: ☐ Case sensitive ☐ Regular exp.

Text

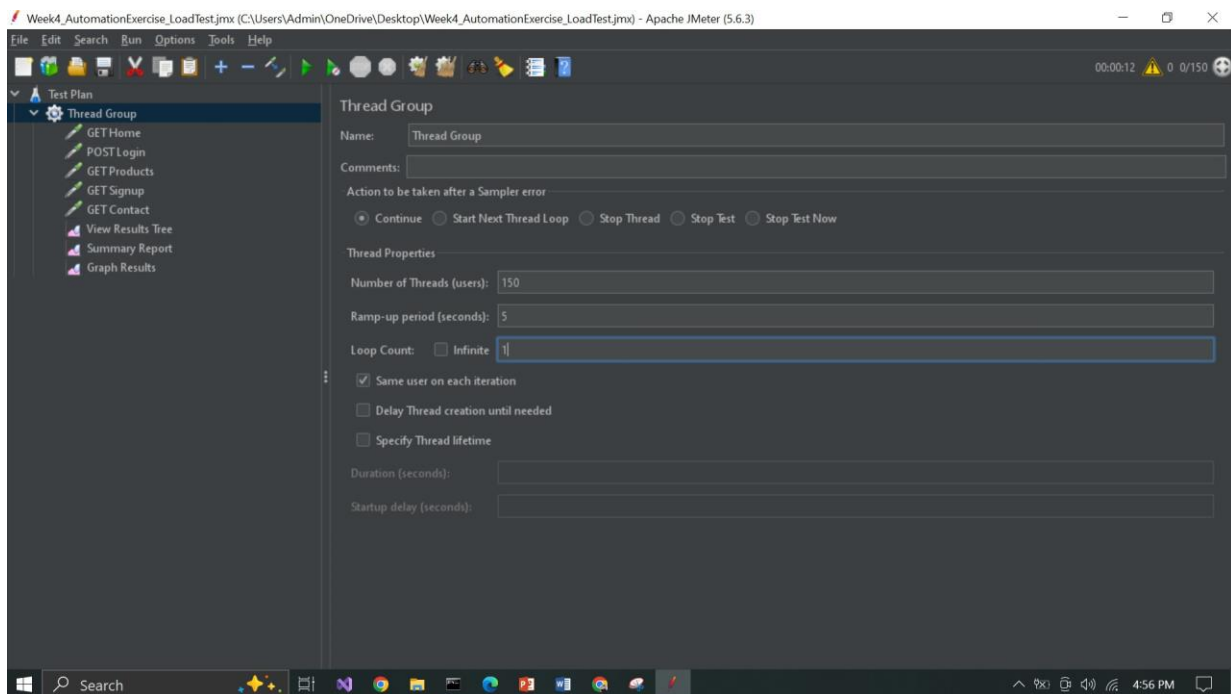
- GET Contact
- GET Home
- POST Login
- GET Signup
- GET Signup
- GET Contact
- GET Home
- GET Contact
- GET Home
- GET Signup
- GET Products
- POST Login
- POST Login
- GET Signup
- GET Contact
- POST Login

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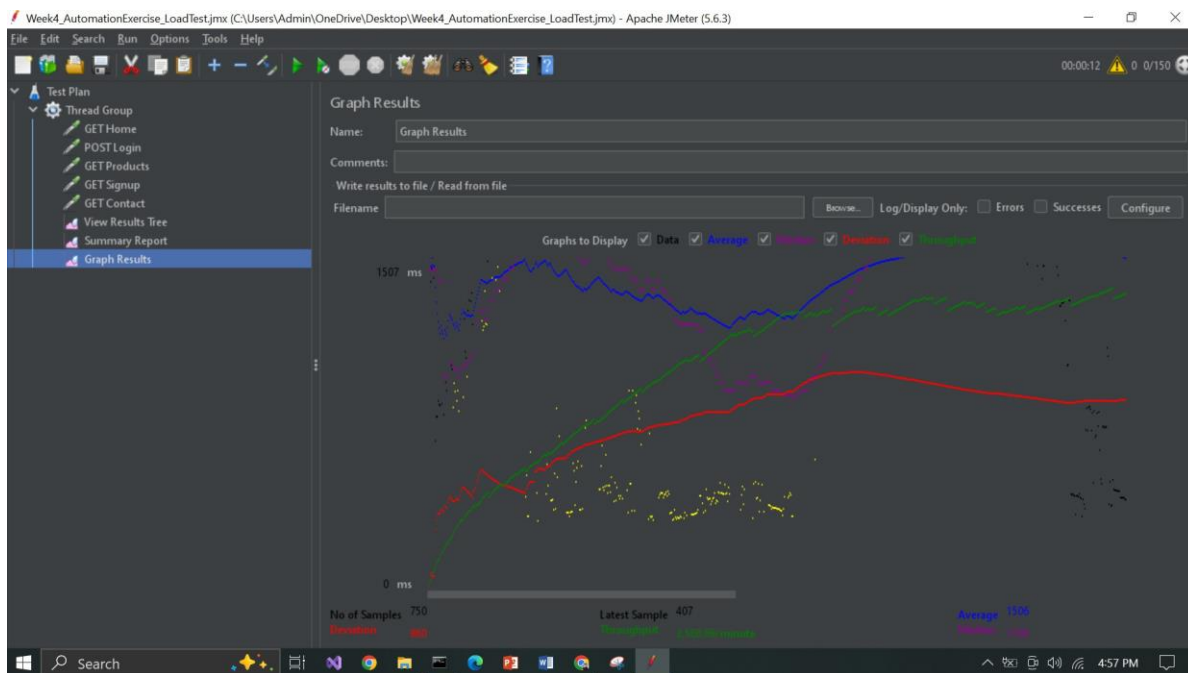
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5.3 – TS-03: Spike Test (150 Users, 5s Ramp-up, 1 Loop)

Thread Group Configuration:



Graph Results – Spike pattern in response time:



Summary Report – Actual JMeter results:

Week4_AutomationExercise_LoadTest.jmx (C:\Users\Admin\OneDrive\Desktop\Week4_AutomationExercise_LoadTest.jmx) - Apache JMeter (5.6.3)

File Edit Search Run Options Tools Help

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Test Plan

- Thread Group
 - GET Home
 - POST Login
 - GET Products
 - GET Signup
 - GET Contact
 - View Results Tree
 - Summary Report**
 - Graph Results

Summary Report

Name: Summary Report

Comments:

Write results to file / Read from file

Filename: Browse... Log/Display Only: ☐ Errors ☐ Successes

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB...	Sent KB/sec	Avg. Bytes
GET Home	150	1819	381	3785	855.62	18.00%	18.7/sec	805.96	3.42	44060.1
POST Login	150	1466	307	2909	858.17	100.00%	17.5/sec	60.16	3.95	3515.1
GET Products	150	1613	313	3103	917.58	33.33%	16.1/sec	518.15	2.12	33052.2
GET Signup	150	1400	313	2993	794.99	28.67%	18.8/sec	108.05	2.44	5890.6
GET Contact	150	1230	329	3675	750.67	14.67%	19.3/sec	164.88	2.58	8742.6
TOTAL	750	1506	307	3785	860.74	38.93%	59.3/sec	1104.14	9.54	19052.1

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View Results Tree – Pass/Fail per request:

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File Edit Search Run Options Tools Help

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Test Plan

- Thread Group
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View Results Tree

Name: View Results Tree

Comments:

Write results to file / Read from file

Filename: Browse... Log/Display Only: ☐ Errors ☐ Successes

Search: Case sensitive ☐ Regular exp. ☐ Search

Text

- GET Products
- GET Signup
- GET Signup
- GET Signup
- GET Signup
- GET Products
- POST Login
- GET Signup
- GET Products
- GET Products
- POST Login
- POST Login
- GET Signup
- GET Products
- GET Products
- GET Products
- GET Signup
- GET Signup
- POST Login

☐ Scroll automatically?

Raw Parsed

6. KPI Pass / Fail Summary

KPI	Actual Value	Result
TS-01 Avg Response Time	735 ms	PASS
TS-01 Overall Error Rate	36.60%	FAIL
TS-01 POST Login Error Rate	100%	FAIL
TS-01 Throughput	21.9/s	PASS
TS-01 Max Response Time	5,056 ms	FAIL
TS-02 Avg Response Time	409 ms	PASS
TS-02 Overall Error Rate	20.00%	FAIL
TS-02 POST Login Error Rate	100%	FAIL
TS-02 Throughput	31.1/s	PASS
TS-03 Avg Response Time (Spike)	1,506 ms	PASS
TS-03 Overall Error Rate	38.93%	FAIL
TS-03 Throughput	59.3/s	PASS

7. Conclusion

The performance testing of **automationexercise.com** revealed a critical issue across all three test scenarios: **POST /login returned a 100% error rate** regardless of the number of concurrent users. This indicates that the login endpoint either rejects automated POST requests (likely due to CSRF token protection, bot detection, or session-based form submission requirements), or the credentials used are not valid for the test environment.

Excluding the POST Login errors, the remaining GET endpoints (Home, Products, Signup, Contact Us) performed within acceptable limits for the Load Test (100 users), with average response times between 661ms and 902ms. Under the Spike Test (150 users, 5s ramp-up), response times rose significantly — averaging 1,506ms overall — indicating the server struggles under sudden traffic bursts.

The Stress Test (200 users) interestingly showed lower average response times than the Spike Test, likely because the 60-second ramp-up gave the server time to adjust, compared to the aggressive 5-second ramp-up of the Spike Test.

8. Recommendations

ID	Recommendation	Details	Priority
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R-01	Fix POST Login for Automated Testing	The application likely uses CSRF tokens. Use JMeter's HTTP Cookie Manager and Regular Expression Extractor to capture and replay CSRF tokens dynamically.	Critical
R-02	Implement Server-Side Caching	Add caching (e.g., Redis/Memcached) for static pages like Home, Products, and Signup to reduce server load under high concurrency.	High
R-03	Add Rate Limiting & Auto-Scaling	Configure auto-scaling on the hosting server to handle sudden spikes. The 5-second ramp-up caused severe response time increases (1,506ms avg).	High
R-04	Optimize Login Endpoint	Even when login works, the POST /login endpoint should be optimised with connection pooling and faster session management.	Medium
R-05	Add Performance Monitoring	Integrate a tool like New Relic or Datadog for real-time performance monitoring in production to catch regressions early.	Medium
R-06	Re-run Tests After Fix	Once CSRF token handling is fixed in JMeter, re-run all 3 scenarios to get clean, accurate error rate data.	High